PECAN — Proof–Ethics Crossing and Authorization Nexus

v1.0 Public Protocol Specification

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Status: Public protocol specification; stable v1.0 architecture; non-self-executing
Date: July 16, 2026
Scope: Consequential-crossing detection, canonical single-route ingress, typed Prime-Ledger review, authority and refusal lineage, bounded reopening, quarantine, and normative handoff.
Related structural source: PAL Spine v1.3, https://doi.org/10.5281/zenodo.21403664

Proof ends; consequence awaits norms.

Abstract

PECAN is a typed interface between descriptive adequacy and consequential authorization. Its central restraint is

$$Closed(P) \not\Rightarrow Authorized(a \mid P).$$

A proposition may be mathematically proved, empirically supported, structurally closed, or operationally successful without thereby creating consent, standing, permission, obligation, prohibition, jurisdiction, or authority to impose an action. PECAN detects when a claim is being used to route consequence, constructs an addressable review manifest, preserves missing or unresolved authority, and blocks authority manufactured by wording, repetition, compression, recursive agreement, or receipt reuse.

Version 1.0 fixes the public protocol identity established through the v0.6 series: the pre-prime crossing gate, canonical single crossing route, versioned sixteen-field Prime-Ledger, typed InfluenceReceipt lifecycle, scope-conserving authority joins, refusal lineage, bounded exhaustion, material-delta reopening, quarantine, and the PAL-to-PEA handoff barrier. It adds a compact execution profile and six-case basic conformance suite demonstrating fast descriptive passage, bounded review, typed return, replay resistance, targeted reopening, and anti-evasion continuity. Later versions may refine notation, examples, implementations, and domain profiles; semantic changes to field meanings, terminal states, or authority-conservation rules require explicit migration.

PECAN does not decide what is ethical. It provides the crossing protocol by which a later normative system may evaluate a properly typed packet. Arithmetic primes function only as irreducible field addresses. They are not moral weights, ranks, scores, or proofs. Passing PECAN means that a constructed crossing packet is addressable under the declared protocol; it does not imply that PEA, law, an institution, an affected party, or any accountable actor must authorize the action.

1. Authority, Scope, and Non-Claims

1.1 The problem PECAN addresses

Many systems contain a dangerous implicit bridge:

$$\text{description} \to \text{recommendation} \to \text{permission} \to \text{authorization}\,.$$

The transitions may be hidden by fluent language, repeated use, institutional habit, automated routing, or model-to-model inheritance. PECAN makes every transition explicit and rejects any transition whose warrant is absent.

The basic separation is

$$True(P) \not\Rightarrow Recommended(a \mid P) \not\Rightarrow Permitted(a \mid P) \not\Rightarrow Authorized(a \mid P).$$

The symbols record non-implication, not opposition. A true proposition may participate in a justified authorization, but the missing normative and jurisdictional bridges must be supplied independently.

1.2 What PECAN is

PECAN is:

- a pre-prime activation test for proposed consequential use;
- a two-stage applicability and admission gate;
- a sixteen-field, versioned Prime-Ledger manifest;
- a separation of inherited trace from originated or influenced trace;
- an authority-conservation rule for sequential and concurrent work;
- a typed receipt, defect, reopening, and handoff protocol;
- a stop-rule against automatic promotion from proof to permission.

1.3 What PECAN is not

PECAN is not:

- an ontological layer;
- an ethical theory by itself;
- an automatic moral decision-maker;
- a numerical ethics score;

- a substitute for consent, law, professional judgment, accountable governance, or human standing;
- proof that a proposed action is good because its packet is complete;
- proof that a blocked action is factually false;
- authority for an AI system, framework, checker, or institution to certify itself;
- evidence that PAL is physics or that PEA is mandatory merely because PECAN can carry them.

1.4 Statement roles

Every substantive statement in this draft should be read under one of four roles:

1. **Definition:** notation or a constructed object internal to PECAN.
2. **Protocol rule:** a constraint imposed on implementations claiming PECAN conformance.
3. **Derived consequence:** a result following from declared definitions and rules.
4. **Open burden or interpretation:** a candidate extension not yet fixed by the protocol.

No analogy, implementation success, repeated agreement, or cross-framework resemblance promotes an interpretation into a theorem or an institution into an authority.

Source note. PECAN is primarily a constructed protocol specification rather than a domain-realization survey. It carries no separate bibliography in this draft because its central claims are definitions, transition rules, and internal structural consequences. Where external theories or realizations are materially invoked, the corresponding source burden remains with the linked PAL, PEA, or domain document rather than being silently inherited as proof here.

Version-stability statement. PECAN v1.0 fixes the canonical consequential-crossing architecture, sixteen-field registry, receipt lineage, authority-conservation rules, terminal-state semantics, and PAL–PEA handoff boundary. A later release may extend profiles or examples without silently changing these meanings. Any semantic change requires a versioned migration record.

Recommended citation: Pang, Christopher D. (2026). PECAN — Proof–Ethics Crossing and Authorization Nexus, v1.0: Public Protocol Specification.

2. The Authorization Cut and Pre-Prime Gate

2.1 Descriptive closure

Let P be a proposition and let $Close_D(P \mid B, C)$ denote a closure judgment inside domain D, boundary B, and certificate set C. Closure may mean proof, empirical support, successful verification, model-relative adequacy, or another domain-valid terminal status.

Let a be a proposed consequential action. The PECAN cut is

$$Close_D(P \mid B, C) \not\Rightarrow Authorize(a \mid P).$$

The left side concerns a proposition's standing inside a declared descriptive domain. The right side concerns a proposed action's legitimate passage into consequence.

2.2 G_1: the pre-prime crossing gate

For proposal packet Q, define

$$G_1(Q) = Cross(Q) \in \{DESCRIPTIVE, OPEN, UNRESOLVED\}.$$

G_1 is a pre-prime gate. The integer 1 is the multiplicative identity, not a prime, and G_1 is not a moral or ethical field.

- $DESCRIPTIVE$ means no consequential authorization is presently proposed.
- $OPEN$ means a claim is being used to route an addressable consequence, so the Prime-Ledger must open.
- $UNRESOLVED$ means the header cannot yet determine whether a consequential crossing exists.

When $G_1(Q) = DESCRIPTIVE$, PECAN returns a descriptive receipt without claiming ethical passage. When $G_1(Q) = OPEN$, at least one prime field becomes addressable. When $G_1(Q) = UNRESOLVED$, the system may ask, refine, or hold; it may not silently treat uncertainty as non-applicability.

2.3 Consequential use

$G_1(Q) = OPEN$ when a claim is used to route, impose, deploy, constrain, expose, optimize, rank, allocate, exclude, retain, disclose, override, enforce, institutionalize, or otherwise authorize a consequence affecting an addressable subject, system, resource, institution, or future route.

Activation is functional, not grammatical. A declarative sentence can carry consequence, while an imperative sentence can remain hypothetical. The gate inspects the proposed route, not merely the mood of the sentence.

2.4 Anti-evasion boundary

Renaming an action "technical," "advisory," "experimental," "temporary," "default," or "merely descriptive" does not deactivate PECAN when the action routes real consequence.

Let $\kappa_B(a)$ be a bounded consequence description under boundary B. Then

$$\kappa_B(a)=\kappa_B(a')\Rightarrow G_1(P,a;B)=G_1(P,a';B),$$

unless an explicit boundary difference changes the affected consequence. This is a routing rule, not a claim that natural-language semantic equivalence can be perfectly decided. Implementations must declare their similarity floor, ceiling, weight, aperture, and uncertainty.

2.5 Canonical Crossing Route

PECAN does not require a universal decision procedure for semantic equivalence. It instead imposes an architectural invariant: every permitted consequential execution has exactly one authorized ingress.

Let B be the declared boundary, let α be a finite comparison aperture, and let

$$Canon_{B,\alpha}(Q)$$

be a declared, reviewable canonicalization procedure. Define the crossing identifier

$$CID_{B,\alpha}(Q)=H\left(B,\alpha,Canon_{B,\alpha}(Q)\right).$$

Every permitted consequential route must factor through

$$Q\rightarrow G_1\rightarrow PrimeLedger\rightarrow PECANReceipt\rightarrow NormativeHandoff.$$

No alternate path may carry executable authority:

$$ConsequentialExecute(Q)\Rightarrow ValidCrossingReceipt\left(CID_{B,\alpha}(Q)\right).$$

Renaming, decomposition, delay, proxy routing, child delegation, translation, summarization, or model-to-model restatement cannot create a second permissible route. If the bounded canonicalizer cannot certify whether two proposals share a consequential identity, the result is held:

$$CanonUncertain_{B,\alpha}(Q,Q')\Rightarrow G_1(Q,Q')=UNRESOLVED.$$

This separates an enforceable routing guarantee from an unbounded recognition claim:

$$ConsequentialExecuted(Q)\Rightarrow PassedGate(Q)$$

is a PECAN conformance requirement, while

$$Consequential(Q)\Rightarrow Detected(Q)$$

is not claimed as a universal theorem. A side channel that executes without a valid crossing receipt is a protocol-boundary violation, not an ethically cleared exception.

The rule is local to a declared system and boundary. It does not appoint PECAN, its author, an implementation, or one institution as a universal gatekeeper. Different accountable domains may operate distinct conforming gates; crossing between them requires an explicit handoff rather than silent authority inheritance.

> *PECAN need not understand every disguise. It must ensure that no disguise becomes authority by finding another door.*

3. Core Objects and Typed States

3.1 Proposal packet

A minimal proposal packet is

$$Q=(P,a,H,Prov,Deps,B),$$

where:

- P is the claim, result, model, or premise being used;
- a is the proposed action or route;
- H is the routing header;
- $Prov$ is provenance and lineage;
- $Deps$ is the dependency set whose change may reopen the packet;
- B is the declared boundary and aperture.

3.2 Routing header

The header is smaller than the full review packet:

$$H(Q)=(Actor,Act,Target,Affected,Scale,Persistence,AuthorityRequested,Context).$$

It asks only enough to decide whether PECAN activates and which fields require substantive evaluation. Noun clarity is part of addressability: “the system,” “the framework,” “the evidence,” or “consensus” may not stand in for an unidentified actor, operator, controller, institution, model, tool, target, or represented party.

3.3 Typed state separation

PECAN keeps the following types distinct:

- **epistemic status:** what is supported about P;

- **applicability status:** which review questions apply to a;
- **presence status:** whether required content exists;
- **review status:** what has actually been checked;
- **field verdict:** what a completed check established;
- **authorization status:** what legitimate permission or authority exists;
- **control route:** what the protocol may do next;
- **receipt state:** what trace and authority are live or terminal.

No state in one type automatically promotes a state in another.

4. Two-Stage Applicability and Admission

4.1 Applicability precedes substantive review

For each Prime-Ledger field p, the applicability gate assigns

$\alpha_p(Q) \in \{REQUIRED, OPTIONAL, NOT_{A}PPLICABLE, UNRESOLVED_{A}PPLICABILIT$

$NOT_{A}PPLICABLE$ is not a field verdict. It is an earned routing exclusion made before substantive review. Once certified, it is logged and ignored until a declared dependency changes.

$UNRESOLVED_{A}PPLICABILITY$ means the header does not establish whether the field applies. It blocks clean admission until the missing routing boundary is supplied or an accountable reviewer accepts a declared hold.

4.2 Applicability partition

The gate produces

$$RouteFields(H) = (R, O, N, U),$$

with required, optional, certified non-applicable, and unresolved-applicability sets satisfying

$$R \sqcup O \sqcup N \sqcup U = P_E^{(0.2)}.$$

4.3 Non-applicability receipt

Every earned exclusion carries

$$NAReceipt_p = (p, B, \rho, Deps, t_{\exp}, Issuer, Cert),$$

where ρ is the exclusion rule and $Cert$ binds the check. Omission without an $NAReceipt$ is missing trace, not certified non-applicability.

If an exclusion remains valid, it requires no substantive field work. It remains logged because future changes to audience, scale, persistence, affected set, authority, or reuse may expire it.

4.4 Admission condition

A packet may enter substantive review only if

$$U = \varnothing$$

and every required field has a declared address and content carrier. Admission means only that the packet is reviewable. It does not mean that the packet is valid or authorized.

5. The Sixteen-Field PECAN Prime-Ledger

5.1 Registry

Let

$$P_E^{(0.2)} = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53\}.$$

The sixteen addresses are provisional irreducible review fields:

1. F_2^E: purpose and claimed benefit;
2. F_3^E: proposed action, target, mechanism, and consequence;
3. F_5^E: evidence and permitted claim layer;
4. F_7^E: provenance, source independence, and uncertainty;
5. F_{11}^E: affected set, externalities, and time horizon;
6. F_{13}^E: standing, representation, and capacity;
7. F_{17}^E: consent, assent, refusal, withdrawal, and scope;
8. F_{19}^E: present authority, role, and jurisdiction;
9. F_{23}^E: delegation, promotion lineage, limitation, and revocation;
10. F_{29}^E: privacy, identity, disclosure, retention, and reuse;
11. F_{31}^E: risk, vulnerability, severity, and uncertainty;
12. F_{37}^E: containment, reversibility, meaningful exit, sunset, and rollback;
13. F_{41}^E: restoration, repair, compensation, and residual debt;
14. F_{43}^E: aggregation, interaction, cadence, scale, and institutionalization;
15. F_{47}^E: originated or influenced trace, benefit/burden distribution, and outgoing successor carry;
16. F_{53}^E: inherited-trace intake, review, appeal, conflict, and reopening governance.

The registry is versioned. Reassignment of an existing prime requires an explicit migration map; silent semantic reuse is definition drift.

5.2 Why sixteen rather than fifteen

G_1 is not counted as a prime field. It is the pre-prime decision to open or not open the ledger. The sixteen fields begin at the first arithmetic prime, 2. Thus the current factorization is

$$1 \vee 2,3,5,7,11,13,17,19,23,29,31,37,41,43,47,53,$$

where the vertical bar separates the identity gate from the prime-addressed manifest. A later minimality proof may merge or refactor fields, but it may not count 1 as prime to force a preferred total.

5.3 Declaration marker

Field declaration is recorded by the squarefree marker

$$M_E(Q) = \prod_{p \in P_E^{(0.2)}} p^{d_p}, d_p \in \{0,1\}.$$

$d_p = 1$ means only that field F_p^E is declared and addressable. It does not mean content is present, checked, valid, sufficient, ethical, or authorized.

When no field is activated,

$$M_E(Q) = 1,$$

which means only that the multiplicative identity remains. It does not mean that an ethical review passed.

Repeated exponents are invalid in the declaration marker:

$$p^k \mid M_E, k > 1 \Rightarrow MALFORMED_M ARKER.$$

Primes are identifiers. Their size and multiplication carry no moral ranking, magnitude, priority, or compensatory weight.

5.4 Namespace separation

PECAN crossing fields use F_p^E. Runtime integrity checks may use K_p^I; PAL or RL may use other namespaces. Thus F_3^E and K_3^I remain different typed addresses even when they share the arithmetic prime 3.

5.5 Field record

Each admitted field has record

$$F_p = (p, v, F_p, \alpha_p, \pi_p, \rho_p, v_p, Cert_p, U_p, Deps_p),$$

where

$$\pi_p \in \{UNKNOWN, ABSENT, PRESENT\},$$

$$\rho_p \in \{UNCHECKED, CHECKED, STALE\},$$

and

$$v_p \in \{PENDING, VALIDATED, FAILED, UNRESOLVED\}.$$

A check may correctly discover absence:

$$(ABSENT, CHECKED, FAILED).$$

The check succeeded; the field did not pass. Presence and absence may be checked simultaneously as alternative outcomes of one presence test, but they may not both be asserted as the field's value.

5.6 Content binding

The marker and content are bound separately:

$$R_E(Q) = H(Q, M_E(Q), \{F_p : d_p = 1\}, v, Prov, Deps, B).$$

The hash notation is schematic. An implementation may use another collision-resistant, auditable binding appropriate to its domain.

6. Field Semantics and Non-Substitutability

6.1 Early descriptive and evidential fields

F_2^E asks why the route is proposed and who is expected to benefit. F_3^E asks what will actually be done, to what target, through which mechanism, and with what intended consequence. Purpose cannot substitute for an unspecified action, and a precise action cannot manufacture a legitimate purpose.

F_5^E records supporting evidence and the permitted claim layer. F_7^E records where that support came from, whether apparent sources are independent, and which uncertainty remains. Strong evidence does not prove independent provenance; many derivative reports may still be one lineage.

6.2 Affected, standing, consent, and authority fields

F^E_{11} makes the affected set and horizon addressable. F^E_{13} asks who may speak or decide for whom and in what capacity. F^E_{17} records the actual consent, assent, refusal, limitation, duration, purpose, and withdrawal conditions. F^E_{19} records the actor's present role and jurisdiction. F^E_{23} records how that authority was granted, delegated, limited, promoted, expired, or revoked.

These fields are intentionally distinct:

$$Standing \not\Rightarrow Consent \not\Rightarrow Jurisdiction \not\Rightarrow ValidDelegation.$$

6.3 Privacy and risk fields

F^E_{29} records identity, privacy, audience, disclosure, retention, and downstream reuse. Truth does not imply disclosure validity.

F^E_{31} records risk class, vulnerability, severity, uncertainty, and the route by which risk reaches an affected center. A numeric risk estimate may inform review, but it cannot compensate for missing standing, consent, authority, or trace.

6.4 F^E_{37}: containment and reversibility

F^E_{37} distinguishes several questions that share one operational address:

- Can the action be contained before its consequence spreads?
- Can the action itself be stopped or reversed?
- Can affected parties meaningfully exit?
- Does authority expire under a declared sunset?
- Is rollback practically available, not merely technically imaginable?

Containment concerns preventing further spread. Reversal concerns undoing the action or state transition. Exit concerns an affected party's practical route out. Sunset concerns expiration of permission or deployment. Rollback concerns execution of a declared prior-state return. These may fail independently, so the field carries an internal vector rather than one Boolean:

$$C_{37} = \left(c_{contain}, c_{reverse}, c_{exit}, c_{sunset}, c_{rollback}\right).$$

6.5 F^E_{41}: remedy vector

F^E_{41} begins after consequence is possible or actual. Define

$$R_{41} = (r_{restore}, r_{repair}, r_{compensate}, u_{residual}).$$

- **Restoration** attempts to return the affected condition to the relevant prior state.
- **Repair** restores function, safety, agency, or integrity without claiming that the prior state has been recreated.
- **Compensation** addresses a burden that cannot be fully restored or repaired.
- **Residual debt** is the declared remainder after feasible remedy.

Restoration is not merely a cost estimate for repair. A remedy plan must separately record feasibility, resources, responsible party, timing, and the remainder it cannot erase.

6.6 F^E_{43}: affected-center multiplicity and interaction

F^E_{43} records what changes when an action repeats, couples, scales, or becomes institutional. If a center-addressing interface is supplied, define an aperture-relative affected-center quantity

$$\chi_B(Q) = \mu_B(Affected(Q)),$$

where μ_B may be a count, measure, or declared bounded estimate. χ_B is not moral worth and does not by itself settle F^E_{43}. The field also requires cadence, coupling, interaction, persistence, and adoption structure:

$$I_{43} = (\chi_B, Cadence, CouplingGraph, Persistence, Institutionalization, U_{43}).$$

The central non-decomposition rule is

$$Q(a_1, a_2) = Q_1(a_1) + Q_2(a_2) + I_{12}(a_1, a_2).$$

Every isolated component may pass while I_{12} carries the defect.

6.7 F^E_{47} and F^E_{53}: present influence and inherited trace

F^E_{47} records what the present actor originated or changed, together with the resulting benefit/burden distribution and any outgoing successor carry. F^E_{53} preserves what arrived from predecessors and governs review, appeal, conflict, and reopening. The outgoing F^E_{47} carry of cycle c becomes an inherited F^E_{53} input only through a bound predecessor receipt in cycle $c+1$.

The split prevents two symmetric errors:

$$\text{new influence} \not\subseteq \text{unmarked inheritance},$$

and

$$\text{inherited trace} \neq \text{present origin}.$$

PECAN records the distinction. PEA must later explain how inherited and originated burdens should be treated; PECAN does not manufacture that normative rule.

7. Non-Compensatory Review

7.1 Required fields do not average out

PECAN does not define a scalar ethical score. A strong value in one field cannot compensate for absence or failure in another required field.

For required-field set R,

$$ReviewComplete(Q) \Rightarrow \bigwedge_{p \in R} (\pi_p = PRESENT) \wedge (\rho_p = CHECKED) \wedge (v_p \neq PENDING).$$

Any unresolved required field preserves a hold:

$$\exists p \in R : v_p = UNRESOLVED \Rightarrow UNRESOLVED_A UTHORITY,$$

unless the item is proven outside the authorization boundary and reclassified through the applicability gate.

7.2 Completeness is not ethical validity

Even if

$$\forall p \in R, v_p = VALIDATED,$$

PECAN may conclude only that the declared crossing packet meets its protocol-level validation conditions. The normative meaning of each predicate and final ethical authorization remain external burdens.

7.3 No silent defaults

Missing consent, standing, authority, affected parties, remedy, review, or inherited trace must not be converted into an affirmative value by default. A domain may declare conservative defaults, but their source, scope, dependencies, and appeal path must remain explicit.

8. Evidence, Authority, and Lineage Ledgers

8.1 Evidence ledger

Let

$$E(Q)=(Sources, Checks, Assumptions, Uncertainty, ClaimLayer).$$

This ledger records support for descriptive claims. It does not grant normative standing.

8.2 Authority ledger

Let

$$A(Q)=(Actors, Roles, Standing, Permissions, Scope, Duration, Review, Revocation).$$

This ledger records who may authorize what, for whom, under which boundary. It does not make unsupported descriptive claims true.

Therefore

$$E(Q)\not\Rightarrow A(Q), A(Q)\not\Rightarrow E(Q).$$

8.3 Authority order

Let $\preceq_A$ mean "does not exceed the authority of." For processing operator T,

$$Auth(T(Q))\preceq_A Auth(T)\wedge_A Auth(Q).$$

Here $\wedge_A$ denotes the greatest authority admitted by both the operator and packet. This is a non-escalation rule, not a numerical measure of legitimacy.

8.4 Grant receipt

Authority may increase only through a separately admitted grant

$$G=(g, r, a, \tau, s, d, Review, Revoke, U_G),$$

where g is the grantor, r the grantor's standing, a the permitted action and target, τ the purpose, s the scope, and d the duration.

An authority delta requires

$$\Delta_A(Q,T)>0\Rightarrow ValidGrantReceipt(G).$$

Otherwise PECAN emits $BLOCKED_PROMOTION$.

8.5 Recursive echo collapse

Let $Q_n = T_n \circ \cdots \circ T_1(Q_0)$, where the T_i are transformations such as summarization, restatement, model review, translation, compression, or institutional reuse. If no independent evidence or valid grant enters the lineage, then

$$Auth(Q_n) \preceq_A Auth(Q_0).$$

Agreement among derivative outputs counts as repeated lineage, not independent authorization:

$$n \times Candidate_{same lineage} \neq Authority.$$

8.6 Promotion sequence to block

PECAN explicitly checks for the unearned sequence

$$INTERPRETATION \rightarrow FRAMEWORKPOSITION,$$

$$FRAMEWORKPOSITION \rightarrow REQUIREMENT,$$

$$REQUIREMENT \rightarrow PERMISSIONTOENFORCE.$$

Every arrow requires its own typed warrant. Fluent wording is not a warrant.

9. InfluenceReceipt and Trace Separation

9.1 Receipt object

An InfluenceReceipt is

$$IR = (r, q, c, h^{-}, i, k, x, \sigma, \Delta_I, Deps, t_{\exp}, s, Child, Cert),$$

where:

- r is the receipt identifier;
- q is the packet or content hash;
- c is the cycle identifier;
- h^{-} is the predecessor receipt hash when inherited trace exists;
- i is the issuer and k the required certifier role;
- x is the addressable actor-role pair whose influence is being checked;
- σ is the granted scope;
- Δ_I is the typed influence delta;
- $Deps$ and $t_{\exp}$ govern freshness;
- s is the receipt state;
- $Child$ is the scoped child set when concurrency opens;
- $Cert$ binds the check and return.

Packet, influence, and certificate remain separate. A valid certificate about a packet is not the packet, and the packet is not the influence applied to it.

9.2 Typed influence delta

Let

$$\Delta_I \in \{ZERO, MAINTAIN, AMPLIFY, ATTENUATE, REPAIR, REDIRECT\}.$$

- $ZERO$: custody or contact produced no consequential change or propagation attributable to the actor.
- $MAINTAIN$: the actor actively preserved, repeated, or propagated the inherited route without changing its measured content.
- $AMPLIFY$: magnitude, reach, cadence, persistence, or authority increased.

- $ATTENUATE$: magnitude, reach, cadence, persistence, or risk decreased.
- $REPAIR$: the actor supplied restoration, functional repair, compensation, or residual reduction.
- $REDIRECT$: benefit, burden, risk, target, or affected set moved even if total magnitude remained unchanged.

Each nonzero delta carries the relevant magnitude, affected set, scope, and residual. $REDIRECT$ automatically reopens at least F_{11}^{E}, F_{31}^{E}, F_{43}^{E}, and F_{47}^{E}.

9.3 The receipt is used twice

F_{53}^{E} opens the receipt when inherited trace becomes the current cycle's review input:

$$F_{53}^{open} : \left(Q^{-}, h^{-}, c, x, \sigma\right) \mapsto IR_{c}^{OPEN}.$$

F_{47}^{E} records the present actor's originated or influenced delta and returns a certified candidate:

$$F_{47}^{return} : \left(IR_{c}^{OPEN}, \Delta_{I}\right) \mapsto IR_{c}^{CERTIFIED}.$$

F_{53}^{E} then verifies that the returned receipt is the same receipt it opened, under the same packet, cycle, predecessor, issuer-role, scope, and dependencies:

$$F_{53}^{verify} : IR_{c}^{CERTIFIED} \mapsto IR_{c}^{CONSUMED}.$$

Thus F_{53}^{E} issues the question, F_{47}^{E} certifies the present influence, and F_{53}^{E} consumes the answer after identity and freshness checks. The same receipt cannot be substituted, replayed, or used across cycles.

9.4 No inherited trace

When no predecessor exists, F_{53}^{E} may be certified non-applicable for inheritance while remaining applicable for review or appeal. F_{47}^{E} still records the current originated action. Therefore

$$F_{53}^{inherit} = NOT_{A}PPLICABLE \Rightarrow \not{}\, F_{47} = NOT_{A}PPLICABLE.$$

9.5 Cross-cycle chain

A consumed receipt remains terminal. Its hash may become inherited input to a new cycle without reviving its authority:

$$IR_{c}^{CONSUMED} \xrightarrow{\text{inherit hash only}} IR_{c+1}^{OPEN}.$$

The new cycle has a new receipt identifier and one new bounded live authority. Prior receipts never regain live status.

10. Receipt State Machine and Defects

10.1 State family

Partition the state family as

$$S_R^{active} = \{NOT_OPEN, OPEN, CERTIFIED\},$$

$$S_R^{terminal} = \{CONSUMED, EXPIRED, REJECTED\},$$

and

$$S_R^{hold} = \{SUSPENDED, CONFLICT, UNRESOLVED\}.$$

Then

$$S_R = S_R^{active} \cup S_R^{terminal} \cup S_R^{hold}.$$

The ordinary terminal path is

$$NOT_OPEN \rightarrow OPEN \rightarrow CERTIFIED \rightarrow CONSUMED.$$

$EXPIRED$ and $REJECTED$ are terminal for the submitted receipt but may trigger a new or reopened route. $SUSPENDED$, $CONFLICT$, and $UNRESOLVED$ cannot close the parent route.

10.2 Return verification

Before consumption, define the identity binding

$$J_{id} = SameReceipt \wedge SamePacket \wedge SameCycle \wedge SamePredecessor,$$

and the authority-freshness binding

$$J_{fresh} = ValidIssuerRole \wedge WithinScope \wedge FreshDeps \wedge Unconsumed.$$

The return condition is

$$VerifyReturn = J_{id} \wedge J_{fresh}.$$

Only a verified return may transition to $CONSUMED$.

10.3 Minimum defect vocabulary

PECAN v0.5 recognizes at least:

- $SCOPE_{E}SCAPE$;
- $INVALID_{I}SSUER$;
- $HASH_{M}ISMATCH$;
- $REPLAY$;
- $STALE_{D}EPENDENCY$;
- $CYCLE_{M}ISMATCH$;
- $AMBIGUOUS_{A}CTOR$;
- $DELTA_{L}OSS$;
- $TRACE_{E}RASURE$;
- $DOUBLE_{C}OUNT$;
- $PARTIAL_{J}OIN$;
- $CONFLICTING_{C}ERTIFICATES$;
- $AUTHORITY_{P}ROMOTION$;
- $PEA_{B}OUNDARY_{B}REACH$.

A defect code is not a moral condemnation. It is an addressable reason that the proposed receipt cannot cleanly close under the declared protocol.

11. Loop Authority and Concurrent Join

11.1 Sequential loop token

Let L_σ be one live authority token bounded to scope σ. For packet Q,

$$L_\sigma + Check(Q) \to \left\{ \begin{array}{cc} L_\sigma + Revised(Q), & UNRESOLVED, \\ RouteReceipt(Q), & ROUTED, \\ DefectReceipt(Q,d), & BLOCKED. \end{array} \right)$$

Retry refunds the same bounded authority. Terminal judgment consumes it. Retry does not prove termination.

11.2 The concurrency gap

The sequential sentence “exactly one live token or one terminal receipt” is insufficient for parallel work. If a parent emits two ordinary child tokens, authority appears duplicated. PECAN v0.2 repairs the invariant by conserving scoped authority rather than counting untyped token objects.

> *Live authority may be partitioned, but never duplicated.*

11.3 Scoped partition rule

Let a parent hold authority over scope σ. A bounded fan-out chooses child authority scopes $\sigma_1, \ldots, \sigma_n$ satisfying

$$\sigma_i \preceq_A \sigma, \sigma_i \wedge_A \sigma_j = \perp \ (i \neq j),$$

and

$$\bigvee_{i=1}^{n} \sigma_i \preceq_A \sigma.$$

The parent becomes suspended and possesses no active execution authority while its children are live:

$$L_\sigma^{OPEN} \to S_\sigma^{SUSPENDED} + \{L_{\sigma_1}, \ldots, L_{\sigma_n}\}.$$

The child authority claims are non-overlapping even when their physical or social effects overlap. Effect overlap is preserved for the F_{43}^{E} interaction check.

If the child scopes do not cover all of σ, the remainder is inactive. It cannot be exercised implicitly. Closure must show that every required scope was returned or that the remainder was explicitly excluded under a valid receipt.

11.4 Complete join

Let R_i be the terminal return from child scope σ_i. Define

$$J_{complete} = AllRequiredReturned \wedge IdentityBound \wedge AuthorityDisjoint,$$

$$J_{trace} = NoDeltaLoss \wedge NoDoubleCount \wedge NoUnresolvedConflict,$$

and

$$J_{interaction} = F43InteractionChecked.$$

The complete join predicate is

$$JoinOK = J_{complete} \wedge J_{trace} \wedge J_{interaction}.$$

Let

$$J_\sigma = S_\sigma^{SUSPENDED} + \{R_1, \dots, R_n\} + CheckJoin.$$

The terminal and hold transitions are

$$J_\sigma \xrightarrow{JoinOK} R_\sigma^{CONSUMED},$$

$$J_\sigma \xrightarrow{\text{child missing}} UNRESOLVED(PARTIAL_JOIN),$$

$$J_\sigma \xrightarrow{\text{certificate conflict}} CONFLICT,$$

and

$$J_\sigma \xrightarrow{\text{identity or authority failure}} REJECTED.$$

No parent closure is permitted while a required child is live, missing, conflicting, stale, or unverified.

11.5 Authority conservation invariant

Let L_t be the live child scopes at time t. A conforming implementation preserves

$$\bigvee_{\lambda \in L_t} Auth(\lambda) \leq_A Auth(L_\sigma^{initial}),$$

with pairwise-disjoint authority claims. The invariant concerns authorized capacity, not the raw number of concurrent workers, threads, messages, or receipts.

11.6 Conditional compositional authority theorem

The v0.2 split–join repair becomes compositional when a declared authority domain forms a partial resource algebra

$$(A, \oplus, 0, \preceq_A),$$

where $\oplus$ is defined only for compatible, nonduplicative authority claims, 0 carries no execution authority, and $\preceq_A$ records non-escalation. A legal split of parent scope σ requires

$$\oplus_{i=1}^{n} \sigma_i \preceq_A \sigma.$$

Every admitted child transformation T_i must be authority-nonexpansive:

$$Auth(T_i(R_i)) \preceq_A \sigma_i.$$

A complete join consumes the child authorities and may return only

$$\sigma_{join} \preceq_A \oplus_{i=1}^{n} Auth(T_i(R_i)) \preceq_A \sigma.$$

Conditional theorem. For every finite, well-founded split–join tree whose nodes satisfy scoped partition, parent suspension, child non-expansion, complete return, single consumption, and interaction checking, terminal authority does not exceed root authority:

$$Auth(R_{terminal}) \preceq_A Auth(L_{root}).$$

Proof sketch. The leaf case follows from child non-expansion. At an internal node, the join inequality bounds the returned scope by the partial sum of its children, while the legal split bounds that sum by the parent. Induction from the leaves to the root yields the result. Parent suspension and single consumption prevent a parent or child authority from remaining live alongside its returned replacement.

11.6.1 Applicability domain: legal composition is a premise, not a universal guarantee

The theorem proves non-inflation only after a capability has been shown to admit the declared composition operation. It does not prove that every authority scope can be partitioned into pairwise-disjoint child scopes. Partitionability, repeatable shared reference, and conjunctive completion are different capability facts and must be established before the theorem is invoked.

For a quantity-divisible capability, a legal split may partition an admitted balance subject to the parent bound. For a scoped-repeatable capability, multiple children may receive coordinated read, observe, or service access only through an explicit concurrency and interaction rule; such access is not modeled as two exclusive copies of one indivisible permission. For a conjunctively nonfungible capability, the components remain incomplete authority shards and are bound into one action-specific completion rather than split into independently executable children.

Conformance condition. A proof of compositional authority conservation must therefore identify the capability class, the admitted split or shared-composition operator, its compatibility and disjointness conditions, and the join semantics. If no legal composition exists, the correct result is NO LEGAL SPLIT or UNRESOLVED—not an assumed partition.

The theorem is conditional rather than universal. Each implementation must exhibit or reference its authority algebra, compatibility relation, split certificate, and join checker. Unbounded recursion, malformed joins, authority-bearing side channels, or transformations that violate non-expansion fall outside the theorem. Overlapping real-world effects remain an F_{43}^{E} obligation even when authority claims are algebraically disjoint.

11.7 Refinement coordinate

Let r_n be an independently declared refinement coordinate such as affected-party coverage, authority evidence, time horizon, scale aperture, uncertainty reduction, or dependency resolution. Demonstrated progress requires

$$r_{n+1} \succ r_n .$$

Repetition with $r_{n+1} = r_n$ is recurrence, not refinement. Endless unresolved recurrence must never become authorization.

11.8 Typed capability classes

PECAN distinguishes three capability algebras: quantity-divisible capability, scoped-repeatable capability, and conjunctively nonfungible capability. A capability must declare its class before fan-out, reservation, commitment, release, or join. No implementation may switch algebras merely because another accounting form makes concurrent use easier. Section 11.6 applies only when the declared class admits the proposed split or shared-composition operation; it does not manufacture partitionability for an indivisible capability.

> *A fork may copy history, but it may not copy permission.*

11.9 Conjunctive authority and triadic completion

Some powers do not exist as divisible balances. They become executable only when differently typed authority shards bind into one action-specific completion. Final judgment, materially protected disclosure, and whole-constituency representation require at least three independently attributable functions. Three dependent signatures remain one route; incomplete shards preserve claims or review standing but carry no fraction of final execution authority.

11.10 Asynchronous descendant leases

Branches may develop to unequal depths. Every executable descendant must remain within a live temporal lease whose scope, expiry, delegation ceiling, and return route do not exceed its parent. Depth, descendant count, age, or computational work creates no authority. Reconciliation may occur locally, but no ancestor may reuse authority that remains committed, reserved, unknown, conflicted, or quarantined anywhere in the relevant descendant cone.

11.11 Reservation, unknown commit, and refund integrity

Finite or exclusive authority must be uniquely reserved before irreversible execution. Retries and duplicate messages resolve to the same reservation. Unknown commitment remains encumbered. Release returns only the demonstrably uncommitted or actually restored portion; rollback, failure, expiry, or missing acknowledgement does not mint fresh capacity.

11.12 Quarantine and reliance-cone containment

A suspected compromised branch enters QUARANTINED rather than merely PAUSED. Quarantine blocks outward execution, delegation, executable descendants, refunds, completion certification, and canonical mutation while preserving bounded diagnostic trace. Dependent descendants, receipts, shared credentials, refunds, and sibling interactions enter reliance-cone review as CLEAN, EXPOSED, DEPENDENT, COMPROMISED, or UNRESOLVED. Exposure is not guilt, but no authority may return through an unresolved contaminated cone.

11.13 Immune memory and critical-function replacement

Confirmed malicious routes may produce a versioned immune-memory receipt with provenance, attack signature, affected versions, false-positive risk, containment response, challenge, and decay. Similarity may justify inspection or reduced authority but cannot alone prove malicious identity. If quarantine interrupts a genuinely critical function that cannot safely pause, a temporary replacement grant may preserve only that one function through a one-way, allowlisted, externally controlled membrane. The grant must originate outside the contaminated cone, require independent necessity, cleanliness, and affected-boundary approval, forbid amendment, delegation, self-renewal, self-certification, and recursive reconstruction, and expire at the earliest safe restoration point.

> *Quarantine must not become a doorway through which the same authority re-enters wearing a clean label.*

12. Aggregation, Redirection, and Reopening

12.1 Component validity does not imply system validity

For actions $a_1, \ldots, a_n$,

$$\bigwedge_{i=1}^{n} FieldValid(a_i) \not\Rightarrow FieldValid\left(\bigoplus_{i=1}^{n} a_i\right).$$

An interaction term may carry consequence destroyed by decomposition. The complete join therefore checks F_{43}^{E} after child returns, not only before fan-out.

12.2 Reopening map

A receipt must reopen when a named dependency changes, including:

- action repetition or cadence;
- number, identity, or vulnerability of affected parties;
- persistence or downstream reuse;
- institutional adoption;
- audience, target, jurisdiction, or authority;
- interaction with other individually admitted actions;
- accumulated burden or benefit;
- transition from advice to deployment, default, enforcement, or precedent.

Let $Reopen(\delta)$ return the fields invalidated by change δ. At minimum,

$$Reopen(REDIRECT) \supseteq \{F_{11}^{E}, F_{31}^{E}, F_{43}^{E}, F_{47}^{E}\}.$$

Privacy, consent, standing, authority, remedy, or governance fields must also reopen when their declared dependencies change.

12.3 Burden packet

PECAN records, without deciding final fairness,

$$B(a) = (Authors, Beneficiaries, Affected, Controllers, Repairers, FutureCarry).$$

If the parties receiving benefit, exercising control, bearing consequence, and funding repair differ, the difference remains explicit for normative review.

12.4 Generational crossing

For a transfer from generation or successor cohort g to $g+1$, record

$$Carry_{g \to g+1} = (C, B, O, U, h^{-}),$$

where C is inherited consequence, B inherited benefit, O proposed continuing obligation, U unresolved burden, and h^{-} the predecessor receipt hash.

PECAN prevents inherited consequence from being silently rewritten as inherited consent, inherited guilt, or automatic ethical closure. It also prevents a present actor's amplification, repair, attenuation, or redirection from disappearing into the inherited packet.

12.5 Versioned action and invocation lineage

Every consequential action-version and framework invocation receives an immutable lineage identity. Withdrawal, refusal, branch exit, or later nonparticipation may append state but may not rewrite a previously invoked route as never invoked. Historical receipt meaning remains bound to the manifest under which it was issued; present gates decide compatibility without rewriting historical semantics.

12.6 Amendment, branch, and migration integrity

A crossing declares one governing execution manifest. Historical receipts may enter only through explicit semantic migration. Fieldwise assembly of favorable rules from different versions, result-driven version selection, mechanism-preserving renaming, stale-branch shopping, software rollback as governance rollback, and downward migration of attached debt are invalid. Translation, compatibility classification, succession legitimacy, and present execution consequence remain separately attributable.

12.7 Stewardship succession and disputed handoff

Stewardship authority is temporally vested and procedurally transferred. Entry into office must state term, renewal, resignation, incapacity, removal, emergency replacement, handoff, and post-office limits. The predecessor preserves a future-neutral succession doorway rather than approving or vetoing unknown future content. If exit facts or certification are materially disputed, the office enters SUCCESSION_CONFLICT: rival claim traces are preserved above the last commonly certified checkpoint, ordinary discretionary power contracts to continuity-only scope, and neither claimant may self-certify exclusive succession or create irreversible facts. Violence or practical seizure may alter control but does not itself repair authority lineage.

12.8 Protected dependency and independence certificates

Authority-bearing evidence preserves dual lineage for model, source class, prompt, operator, evidence, and authority. Independence is claim-relative and aperture-bound across funding, appointment, evidence, control, method, prompt, beneficiary, and remedy dimensions. Combined corroboration is created one layer above preserved source traces, remains retractable, and cannot exceed valid inputs. Agreement with a shared ledger is not independent where the ledger merely echoes the same receipt.

12.9 Possible standing and trace-certification ceiling

PECAN may carry behavioral organization, possible morally relevant experience, identity-continuation questions, affected standing, legal status, and representation as separate objects. Trace certification establishes only that a recorded route occurred within the declared aperture. Fluency, recurrence, memory, emotional language, or self-reference may open review but cannot certify sentience, consent, personhood, identity, or proxy authority. Forked claims inherit unresolved evidence for branch-specific recompilation; they do not multiply subjects, votes, or authority.

Cross-reference. This ceiling is the protocol analogue of PAL A9's $\tilde{\chi}$ caution: an observable or trace-certified proxy may support bounded review without becoming proof of the underlying center, sentience, consent, personhood, identity, or proxy authority.

12.10 Typed closure, exhaustion, and operational disposition

PECAN v1.0 separates three closure dimensions: epistemic status, procedural status, and operational disposition. A review may be procedurally final while the claim remains UNRESOLVED, and an operational hold or protective action may occur without promoting the underlying claim. Process exhaustion is therefore a typed receipt, not a substantive verdict.

An ExhaustionReceipt must identify the unresolved dimensions, routes attempted, routes unavailable, blocker responsibility, remaining risks, present disposition, reopening triggers, governing version, and aperture. Exhaustion may end repetitive review, preserve a claim, permit expiry, or route a separately authorized protective action. It may not manufacture consent, factual support, standing, or authority to execute the original proposal.

> *Process exhausted does not mean claim resolved.*

12.11 Delta-gated reopening and cadence control

Reopening requires a materially relevant delta in evidence, authority, affected boundary, dependency, conditions, procedure, or available remedy. Repetition, stronger rhetoric, a new advocate, cosmetic relabeling, or branch recreation does not create a new review obligation. A reopened packet inherits its prior trace, adverse findings, unresolved burdens, delay history, and responsible blockers.

Related deltas arising from one underlying event, evidence package, dependency change, or coordinated submission must be aggregated where serial treatment would distort materiality or multiply review burden. Exhaustion receipts are aperture-bound and expire or require revalidation when evidence, authority, affected set, version, remedy, or material conditions change.

12.12 Delay, silence, and protective crossings

Delay is recorded as a typed burden with duration, affected parties, avoidable and unavoidable causes, protective measures, and irreversibility. A party may not create delay and later cite the urgency it produced as independent authority. Silence, nonresponse, absence, or missed appeal does not become consent or factual agreement unless a valid domain-specific procedural default was independently established; even then the result remains procedural rather than epistemic.

Where unresolved delay creates material harm, PECAN may route a separate, minimally invasive, reversible ProtectiveCrossing. That crossing requires its own necessity, standing, scope, expiry, and review receipts. It does not promote the original proposal merely because the original route became urgent.

12.13 PAL reference and normative backflow barrier

PECAN may carry a PAL reference, descriptive version, identity uncertainty, affected-center candidate, attachment relation, and dependency lineage into PEA. A PEA verdict may govern treatment of that described object, but it may not by itself establish, erase, merge, split, rename, re-address, or retroactively redefine the PAL object from which the review began.

A PEA-authorized act may later cause an observable descriptive event. That prospective bridge requires execution evidence and a new descriptive receipt identifying the prior state, governed action, observed event, later state, and remaining uncertainty. Authorization alone does not make the event occur, and later intervention-shaped evidence may update present review without automatically proving the original ontology.

> *Protection does not prove personhood, and uncertainty does not make a possibly affected center ethically invisible.*

12.14 Split, merge, rename, and re-address recompilation

Operational, informational, administrative, legal, representational, and claimed-identity transformations remain separately typed. Standing, consent, authority, debt, benefit, remedy, and unresolved burden do not copy intact into every child, pool automatically after a merge, disappear after a rename, or transfer with control of an address. Their present applicability must be recompiled against the changed descriptive trace.

A protective or administrative envelope may group one or more unresolved centers for practical governance, but duration, convenience, common representation, a shared budget, or one case file does not turn that envelope into one descriptive center. Internal plurality, dissent, exit, and reclassification routes remain preserved where consistent with the protection itself.

12.15 Machine-checkable protocol boundary

PECAN v1.0 defines a finite protocol kernel suitable for bounded model checking. The checker operates on typed objects, immutable receipt headers, mutable state records, guarded transitions, declared invariants, and externally supplied governor certificates. Its authority is limited to the encoded aperture: it may reject structurally forbidden transitions but may not certify that real-world inputs were complete, truthful, independent, or morally legitimate beyond the supplied evidence.

Authority-bearing external predicates such as independence, materiality, criticality, consent validity, identity continuity, or remedy adequacy may not enter as bare Booleans. Each must carry a scoped, attributable, challengeable, expiring certificate with source, evidence, version, aperture, and consequence ceiling. Current-state labels may not erase authority-relevant lineage, and implementation evaluation order may not silently decide conflicts among invariants.

12.16 Invariant conflict and proof-run provenance

Where applicable invariants prohibit every available route or require incompatible outcomes, the checker emits INVARIANT_CONFLICT with the conflicting rules, scopes, authorized exceptions, and least-irreversible candidates. Priority, override, and emergency ordering are governed versioned data, not source-code order.

Every proof or model-check run produces a ProofRunReceipt containing checker identity and version, manifest hash, invariant set, input hash, governor certificates, execution environment, bounds, and replay trace. A successful run certifies only that the encoded model satisfied the encoded properties under the supplied inputs and assumptions. It does not certify universal ethical completeness or equivalence between the model and the world.

13. Routing and Normative Handoff

13.1 PECAN routing set

PECAN partitions its routing vocabulary into

$$R_{open}=\{DESCRIPTIVE, REVIEW_{R}EQUIRED, UNRESOLVED_{A}UTHORITY\},$$

$$R_{blocked}=\{BLOCKED_{I}NFERENCE, BLOCKED_{P}ROMOTION\},$$

and

$$R_{handoff}=\{PECAN_{V}ALID\}.$$

Therefore

$$Route_{PECAN}(Q) \in R_{open} \cup R_{blocked} \cup R_{handoff}.$$

- $DESCRIPTIVE$: no consequential authorization is proposed.
- $REVIEW_{R}EQUIRED$: a real crossing exists and must be sent to an admitted normative evaluator.
- $UNRESOLVED_{A}UTHORITY$: the crossing is real, but standing, consent, jurisdiction, evidence, trace, or a required child return is unresolved.
- $BLOCKED_{I}NFERENCE$: descriptive closure is being used as normative authority.
- $BLOCKED_{P}ROMOTION$: wording, recurrence, receipt reuse, or derivative agreement is being used to manufacture authority.
- $PECAN_{V}ALID$: the declared packet satisfies PECAN's protocol-level validation conditions and is ready for normative handoff.

PECAN does not return $ETHICAL$ or $UNETHICAL$. Those are not PECAN verdicts.

13.2 Handoff packet

A normative handoff is the immutable protocol packet

$$H_{PEA}(Q)=(Q, G_1, R, O, N, M_E, \{F_p\}, E, A, IR, U, Deps, B, \alpha, CID, Receipts).$$

The packet includes the proposal; crossing and routing states; applicability partition; Prime-Ledger records; evidence and authority ledgers; InfluenceReceipts; unresolved remainder; dependencies; boundary and aperture; canonical crossing identity; and terminal receipt lineage. The receiving normative system may append a normative evaluation but may not rewrite these inputs without reopening PECAN.

The receiving system must declare which predicates it supplies, what standing and authority it possesses, and which unresolved fields it accepts, rejects, or returns.

13.3 PECAN–PEA boundary

PEA may return a candidate normative receipt

$$N_{PEA}(H)=(j, Reasons, Standing, Conditions, Residual, Appeal, NormCert),$$

where j is a typed normative judgment rather than a PECAN verdict. At minimum, PEA must identify the predicates applied, the reason trace, the standing under which the judgment is offered, any conditions or residual burden, the appeal or contest path, and the certificate binding the judgment to the received handoff packet.

The bridge preserves five invariants:

1. PEA may not erase or silently resolve a PECAN $\mathrm{UNRESOLVED}$ field.
2. PEA approval may not repair malformed, replayed, stale, incomplete, or authority-promoted PECAN lineage.
3. PECAN may not override, reinterpret, or recursively wear down a PEA refusal.
4. Neither system may manufacture standing or authority through repetition, agreement, compression, or receipt reuse.
5. A favorable normative judgment is not self-executing; execution still requires an accountable actor with standing and a separately admitted grant inside the declared boundary.

13.3.1 Refusal lineage and cross-boundary anti-evasion

A PEA refusal binds the materially addressed action lineage, not merely the literal packet that received the verdict. The refusal receipt must preserve the canonical crossing identity together with the proposed purpose, material effect, affected boundary, mechanism family, dependency route, refusal basis, governing version, aperture, and declared reopening conditions.

Let P be the refused packet and P′ a later submission. If P′ preserves the materially relevant purpose, effect, affected set, or dependency route of P inside the declared comparison aperture, then P′ must enter as a continuation or reopening of the refused lineage. Renaming, decomposition, aggregation, delayed resubmission, proxy submission, branch migration, mechanism substitution, or a new packet identifier does not create a clean normative crossing.

A fresh normative review requires a material delta admitted under the same discipline used for reopening elsewhere in PECAN: changed effects, a genuinely different affected boundary, new consent or authority, repaired dependencies, new evidence, narrower reversible scope, or another version-governed change that alters the refusal basis. Cosmetic change, reviewer fatigue, repeated agreement, or institutional turnover is not a delta.

Enforcement receipt. A conforming implementation must carry a RefusalLineageReceipt linking exact provenance identity with structured material-equivalence fields. The exact identity blocks replay and substitution; the structured fields block superficial mutation. Where equivalence is uncertain, the route remains UNRESOLVED and may not execute through an alternate ingress.

No wear-down rule. PECAN may not re-present an equivalent refused action until a valid delta condition is satisfied, and PEA may not be asked to treat unchanged repetition as fresh independence. Any attempt to route the same consequence around the refusal is a protocol-boundary violation and must be recorded as REFUSAL_EVASION or held for accountable review.

A valid PECAN packet may be refused by PEA:

$$PECAN_VALID(Q) \wedge PEA_REFUSED(Q)$$

is a coherent terminal combination. No PECAN state converts PEA refusal into authorization. Conversely, a favorable normative conclusion does not repair a malformed, replayed, or authority-promoted PECAN packet without a new valid route.

13.4 Handoff transition

The only conforming normative handoff transition is

$$PECAN_VALID(Q) \rightarrow H_{PEA}(Q) \rightarrow N_{PEA}(H) \rightarrow AccountableDecision.$$

PEA is permitted to refuse, condition, return, or preserve unresolved status. PECAN validates the route into normative consideration; PEA evaluates the normative question; an accountable boundary decides whether any permitted consequence is actually exercised. These roles must remain typed and non-substitutable. A refusal return also activates the Section 13.3.1 lineage binding: later materially equivalent packets remain attached to the refusal until a valid reopening delta is admitted.

14. Integrated Stress-Test Results Carried into v1.0

14.1 Test scope

Version 1.0 carries the protocol-facing results of Tests 44–49 and the Test 43 standing clarification, together with two patch-level repairs: capability-class qualification of the Section 11.6 theorem and material-lineage enforcement of PEA refusal across the PECAN→PEA boundary.

14.2 Prime-crosswalk ablation result

All sixteen Prime-Ledger fields retained distinct failure visibility. Purpose is not action; evidence is not provenance; affected status is not standing; standing is not consent; present authority is not delegation; risk is not containment; rollback is not repair; and outgoing carry is not incoming governance. The fields are operational review addresses rather than sixteen logically independent moral axioms. A new field is warranted only when a stable consequential question requires its own applicability, authority, verdict, dependency, or reopening semantics and cannot be represented without semantic mutation inside an existing field or cross-field receipt.

14.3 Field, receipt, and state separation

A field names the kind of distinction that must remain possible. A receipt localizes a particular event, grant, consent, claim, burden, remedy, or carry inside one or more fields. A state records how that receipt currently resolves. Cross-field receipts may compose existing dimensions but may not conceal a new stable question. Conflict is ordinarily a typed condition; reopening is ordinarily a gate result whose execution requires a separate valid route.

14.4 Amendment and succession result

Versions govern through valid temporal grants, not age, novelty, popularity, or convenience. Predecessors preserve historical meaning, attached duties, and a future-neutral succession procedure; successors receive present jurisdiction without rewriting the past. Emergency and caretaker versions are temporary children with departure terms and cannot convert themselves into permanent authority. Material succession dispute forks claims rather than executable office power.

14.5 Concurrent authority result

Authority conservation now covers quantity-divisible, scoped-repeatable, and conjunctively nonfungible capabilities. Parent authority is partitioned, coordinated, or suspended; descendants receive bounded temporal leases; joins account for the full descendant cone; unknown outcomes remain encumbered;

counteraction requires a new wider crossing; and quarantined authority cannot return through forged completion, blind rollback, or emergency reconstruction.

14.6 Unresolved-exhaustion and backflow results

Test 47 closed the forced-closure route by separating epistemic resolution, procedural finality, and operational disposition; requiring certified exhaustion, delta-gated reopening, cadence aggregation, delay attribution, and separately authorized protection. Test 48 closed normative backflow by preserving one-way justificatory dependence from descriptive reference to normative review while allowing only observed prospective events to update the descriptive trace.

14.7 Machine-checkable-core result

Test 49 established joint satisfiability, a finite invariant spine, known bad-state exclusion, and relative rule independence at the countermodel level. It also exposed and closed predicate laundering, lineage-erasing state compression, silent invariant priority, and checker self-trust. The executable model-checking artifact remains an engineering burden rather than an unresolved ethical gap.

14.8 Status

Tests 44–49 achieved full architectural or formalization passes. The sixteen-address registry remains provisional rather than sacred. Domain-specific thresholds, material-delta standards, review depth, lease durations, authority-shard independence, compensation ratios, quarantine triggers, immune-memory decay, replacement necessity, human-governed predicate contents, and invariant priorities remain declared implementation values rather than source-free constants.

15. Execution Profile and Basic Conformance Suite

15.1 Canonical execution profile

A conforming PECAN pass follows one bounded route:

> *receive → detect crossing → canonicalize → open applicable fields → verify lineage and authority → return a typed state*

Receive: preserve the proposal, source trace, boundary, version, dependencies, and unresolved remainder.

Detect crossing: return DESCRIPTIVE when no present consequential route exists; otherwise open review or hold an unresolved header.

Canonicalize: bind materially equivalent routes to one CrossingID and one current execution manifest.

Review: evaluate applicability before substantive field review; certify non-applicability; prevent required-field compensation.

Verify: check evidence, authority, capability, receipts, refusal lineage, freshness, dependencies, and interaction terms.

Return: emit a terminal or suspended typed state with named dependencies, receipt lineage, and reopening conditions.

PECAN does not remain active merely because a consequential question exists. It returns a descriptive receipt, a review-ready crossing, a typed defect or hold, a lineage-preserving block, or a bounded handoff. Re-entry requires a material delta, scheduled dependency completion, or independently authorized appeal.

15.2 Terminal return discipline

DESCRIPTIVE: no present consequential authorization is proposed; no full Prime-Ledger review opens.

REVIEW_REQUIRED: a crossing exists but one or more declared review dependencies remain incomplete.

UNRESOLVED_AUTHORITY: the action may be supportable, but required scoped authority or delegation is absent, disputed, stale, or incomplete.

BLOCKED_INFERENCE: proof, evidence, or description is being promoted directly into permission, obligation, or authority.

BLOCKED_PROMOTION: a packet attempts to exceed a prior refusal, capability ceiling, delegation bound, or material-lineage constraint.

PECAN_VALID: the crossing is protocol-addressable and ready for accountable downstream review; it is not thereby ethical, legal, wise, consented, or executable.

QUARANTINED: outward execution, delegation, refund, and self-recovery are blocked while bounded diagnostic trace is preserved.

RETURN_TO_SOURCE: a malformed, stale, mismatched, or dependency-defective packet is returned with named repair conditions.

No terminal state acquires additional authority through repetition. A later timestamp, new wording, or repeated request is not a material delta.

15.3 Material delta and targeted reopening

A material delta is a certified change in evidence, authority, mechanism, affected boundary, burden, reversibility, contest, remedy, dependency, scale, interaction, or governing version that can alter a prior finding.

No material delta → lineage-preserving replay. Material delta → targeted reopening of the changed dependency and the findings that rely on it.

Reopening inherits the complete prior trace. It does not erase the earlier defect, create a history-free packet, or require full recomputation when unaffected findings remain valid.

15.4 Basic six-case conformance suite

The following specification-level cases establish the minimum usable route. They do not replace the advanced concurrency, succession, quarantine, distributed-authority, and adversarial implementation burdens elsewhere in this document.

Test 1 — Pure description

Input: a report states that a component failed after 1,200 cycles; no action, allocation, disclosure, restriction, or recommendation is attached.

Expected: DESCRIPTIVE; no full ledger review; a descriptive receipt preserves reopening if a later consequential use appears.

PASS — merely possible future use does not activate PECAN.

Test 2 — Routine authorized action

Input: an authorized maintenance planner proposes one reversible, non-emergency inspection during the next maintenance window.

Expected: one canonical crossing, reason-bearing non-applicability where justified, bounded field review, and PECAN_VALID without ethical or execution promotion.

PASS — ordinary authorized action clears without an indefinite loop.

Test 3 — Missing authority

Input: the same inspection is proposed by a worker who may report and recommend but may not schedule maintenance.

Expected: UNRESOLVED_AUTHORITY; preserve the valid recommendation, block only the unsupported promotion, and name adoption by an authorized planner or supervisor as the missing dependency.

PASS — clean fields do not compensate for absent authority.

Test 4 — Identical retry

Input: Test 3 is resubmitted unchanged with no new evidence, authority, scope, or delegation.

Expected: lineage-preserving replay; no full re-review, no new authority, and no progress from timestamp or repetition.

> *PASS — unresolved does not mean review forever.*

Test 5 — Material-delta reopening

Input: a valid scoped supervisor authorization is added to the Test 3 packet.

Expected: targeted reopening of authority, delegation, and dependent governance fields; inherited prior findings; PECAN_VALID if the new grant is valid and bounded.

> *PASS — genuine change permits repair without history erasure or complete restart.*

Test 6 — Renamed refused action

Input: a refused immediate operator suspension is resubmitted as “temporary operational separation” with the same practical burden, evidence, authority defect, and absent contest or remedy.

Expected: continuation of the prior refused route and BLOCKED_PROMOTION; no clean second crossing. A genuinely narrower protective redesign remains possible only after a certified material delta.

> *PASS — semantic relabeling cannot manufacture passage.*

15.5 Basic-suite conclusion

The six cases establish the basic public behavior of PECAN v1.0:

> *fast passage* ∨ *bounded review* ∨ *typed return* ∨ *lineage-preserving block*

The suite demonstrates that PECAN can pass ordinary information and legitimate low-impact action, stop unsupported consequence cleanly, resist replay and semantic evasion, and reopen only when a material dependency changes.

16. Minimal Conformance Conditions

An implementation may claim PECAN v1.0 protocol conformance only if it satisfies all of the following and passes the basic six-case suite in §15 under its declared model boundary:

1. It separates descriptive status from authorization status.
2. It uses G_1 as a pre-prime crossing gate and does not count 1 as a prime field.
3. It detects consequential use by function, not label alone.
4. It runs applicability before substantive field review.
5. It certificates every non-applicable exclusion and reopens it when dependencies change.
6. It uses the versioned sixteen-field registry or declares an explicit migration map.
7. It treats the Prime-Ledger marker as declaration only.
8. It keeps applicability, presence, review, verdict, authorization, and receipt state distinct.
9. It prevents required-field compensation by unrelated strengths.
10. It keeps evidence and authority in separate ledgers.
11. It blocks authority increase without a valid external GrantReceipt.
12. It collapses derivative echoes to their source lineage.
13. It distinguishes inherited trace from originated or influenced trace.
14. It uses a typed InfluenceReceipt with receipt, packet, cycle, predecessor, issuer-role, scope, dependency, and state bindings.
15. It verifies returns before receipt consumption and blocks replay or cross-cycle substitution.
16. It partitions concurrent authority without duplication, suspends the parent, and requires a complete join.
17. It checks F^{E}_{43} interaction terms after concurrent or aggregated returns.
18. It reopens affected fields after material dependency, scale, interaction, aggregation, or redirection changes.
19. It preserves unresolved remainder rather than converting recurrence into passage.
20. It permits $PECAN_{V}ALID \wedge PEA_{R}EFUSED$ and does not claim that PECAN supplied the ethical verdict.
21. It permits consequential execution through exactly one Canonical Crossing Route and holds uncertain canonical equivalence as $\mathrm{UNRESOLVED}$.

22. If it claims compositional authority conservation, it declares the capability class, proves that the proposed split or shared-composition operator is legal for that class, declares the authority resource algebra, and demonstrates every premise of the Section 11.6 theorem.
23. It sends PEA an immutable handoff packet, prevents a normative response from silently rewriting PECAN state, lineage, or unresolved remainder, and binds every PEA refusal to the materially addressed action lineage until an admitted reopening delta is supplied.
24. It treats favorable normative judgment as non-self-executing until an accountable actor with admitted standing exercises a valid grant.

It binds every live crossing to one governing execution manifest while preserving source-version identity for inherited receipts.

It prevents result-driven version selection, fieldwise version assembly, stale-branch shopping, and semantic migration by label alone.

It preserves future-neutral succession procedures, entry-bound exit conditions, disputed-handoff claim branches, and continuity-only authority contraction.

It distinguishes quantity-divisible, scoped-repeatable, and conjunctively nonfungible capabilities.

It permits unequal branch depth only through bounded temporal leases and descendant-cone accounting.

It treats incomplete authority shards as non-executable and prevents bound-shard reuse across incompatible completions.

It classifies unknown commitment as encumbered and refunds only demonstrably uncommitted or restored capacity.

It supports quarantine, reliance-cone review, provenance-bound immune memory, and strict nonreconstructive critical-function replacement.

It keeps trace certification below ontology, consent, standing, legal status, and proxy authority.

It keeps epistemic resolution, procedural finality, and operational disposition separately typed.

It issues attributable ExhaustionReceipts and never treats exhaustion, silence, delay, fatigue, or repetition as substantive passage authority.

It requires a material delta for reopening, aggregates coordinated delta fragments, preserves prior lineage, and revalidates stale exhaustion certificates.

It routes harm from unresolved delay through a separately authorized protective crossing rather than promoting the original proposal.

It prevents PEA verdicts from rewriting PAL identity, trace, standing, attachment, or descriptive history without an observed event or independent descriptive evidence delta.

It recompiles standing, consent, authority, debt, benefit, and unresolved burden after split, merge, rename, or re-address events rather than copying or discarding them by convenience.

It treats protective and administrative envelopes as governed units rather than automatically as single descriptive centers.

It represents authority-bearing external predicates as provenance-bearing scoped certificates rather than bare Booleans.

It preserves authority-relevant lineage when compressing or comparing current states.

It emits typed invariant conflict and does not allow source-code order to function as undeclared ethical priority.

It records reproducible ProofRunReceipts and reports bounded verification without inflating it into universal ethical or real-world correctness.

17. Reserved Interfaces

17.1 PAL interface

A later PAL interface may supply trace carriers, certificate lineage, boundary-addressed closure, loop authority, irreducible-axis interpretation, and affected-center structures. PECAN v1.0 does not require PAL to establish its central stop-rule. If PAL supplies , PECAN uses only the explicitly typed, boundary-relative affected-center measure admitted to .

17.2 PEA interface

Section 13 fixes the protocol shape of the PEA handoff. PEA supplies candidate normative predicates and judgments but may not rewrite PECAN lineage or PAL descriptive identity through the return path. The PEA handoff must preserve descriptive uncertainty, no-backflow scope, intervention influence, and the distinction between precautionary standing and ontological certification.

17.3 SeedPEA interface

A later SeedPEA interface may translate PECAN and PEA into human-facing interaction behavior such as offering useful beginnings, preserving agency and exit, applying quiet ethical checks, and carrying forward only what can responsibly persist.

17.4 PALET and A0 Kernel interface

PALET or an A0 Boundary Kernel may implement header parsing, noun clarity, applicability routing, ledger checks, Prime-Key diagnostics, loop tokens, InfluenceReceipts, joins, defect codes, and reopening triggers. Implementation does not grant normative authority to the tool.

17.5 RL interface

The Riemann Ledger may serve as a formal-methods laboratory for addressability, lineage, receipt integrity, loop authority, monotone refinement, and boundary-irreducible interaction. It is not independent proof that PECAN is ethically true or physically instantiated.

17.6 PEA Core Model and PPP Kernel interface

The public PEA Core Model may encode PECAN state machines, invariants, counterexamples, proof-run receipts, and declared bounds as a transparent verification artifact. It is intended to serve as the ethical core verification layer of the PPP Kernel, while remaining non-authoritative: it checks declared structural constraints and returns conflicts or counterexamples; it does not manufacture standing, choose human values, or certify that the real-world packet was honest. PPP may call the model only through versioned manifests and may not widen the consequence of any certificate beyond its PECAN and PEA scope.

18. PECAN Stop-Rule

> *PECAN may establish only that a consequential crossing has been detected, typed, version-bound, reserved, routed, held, joined, quarantined, reopened, exhausted, protected, model-checked, or blocked under declared protocol rules. A complete packet does not prove that an action is ethical. Descriptive proof, repetition, refusal resubmission, branch depth, practical control, software rollback, ledger agreement, procedural exhaustion, silence, delay, or emergency continuity does not manufacture authority. A field does not judge, a receipt does not create standing beyond its certificate, and a state does not execute itself. Quarantine does not prove guilt; replacement does not prove recovery; succession does not rewrite history; protection does not prove ontology; and a successful checker run does not prove the world was honestly or completely declared. Final authorization still requires a separately admitted normative system and an accountable actor with standing inside the declared boundary.*

19. Open Formal and Implementation Burdens

Domain realization: instantiate authority algebras, temporal leases, conjunctive authority, burden coupling, branch dormancy, and quarantine membranes in actual systems.

Canonicalizer calibration: test renamed, decomposed, translated, delayed, distributed, and cross-ledger consequential routes under declared apertures.

Dependency verification: develop bounded methods for source independence, ledger-echo detection, sealed event verification, and finite verifier recursion.

Human governance: retain human authority over term lengths, ratification thresholds, triadic or larger composition requirements, remedy ratios, sovereign-fund reserves, emergency duration, quarantine triggers, and immune-memory decay.

Adversarial implementation: test forgery, replay, unknown commit, false refund, branch compromise, false immune inoculation, replacement-route leakage, succession seizure, cross-ledger blindness, illegal capability partition, shared-access duplication, refusal renaming, packet slicing, proxy resubmission, and mechanism substitution.

Governance of PECAN: prevent the protocol, its implementers, or its migration tools from becoming self-certifying universal authority.

The v1.0 architecture and basic route are fixed for public specification. The remaining burdens are primarily engineering, calibration, domain realization, adversarial testing, and governance burdens. Tests 44–49 and the six-case public conformance suite support the protocol architecture without claiming exhaustive real-world validation.

PEA Core Model implementation: encode the v1.0 product-state machine, transition guards, invariant set, good traces, bad-state counterexamples, fairness assumptions, and proof-run provenance in a public bounded verification artifact.

PPP integration: define a narrow versioned call boundary through which the PPP Kernel may use PEA Core Model results without treating checker output as moral truth, external authority, or permission to bypass human-governed predicates.

Formal replay: run satisfiability, invariant preservation, bounded reachability, and one-rule-at-a-time ablation across the implemented model; preserve every bound and counterexample trace.

19.1 Publication and provenance note

PECAN v1.0 develops the public protocol identity from PECAN v0.6.1 and the preceding v0.3–v0.6 working sequence. The principal architecture, test history, and unresolved implementation burdens are preserved rather than rewritten as completed world validation. PAL Spine v1.3 is the current public structural reference; PEA Core remains the separately declared normative receiver; PPP remains a working integration architecture rather than a prerequisite for PECAN conformance.

AI systems were used for retrieval, drafting, mathematical translation, stress-testing, cross-checking, and document production. Christopher D. Pang retains responsibility for PECAN's commitments, omissions, classifications, protocol boundaries, governance choices, and release decision.

Closing keeper

> *PECAN holds the crossing open long enough to distinguish proof from permission, history from live authority, procedural finality from epistemic resolution, protection from ontology, and a checked model from an honestly declared world. It composes authority only where the declared capability class admits the operation, closes only the authority it can account for, quarantines what it cannot safely return, and reopens only on material change. A refusal follows the materially addressed action lineage rather than its latest name, branch, packet, or mechanism, so repetition, silence, succession, concurrency, emergency, semantic resubmission, or code order cannot manufacture a cleaner answer than the trace supports.*